

(1) REINFORCE

Policy Network $\pi_{\theta}(a|s)$

Maximize

$$J(\theta) = \mathbb{E}_{z \sim \pi_{\theta}} r(z)$$

$$z = (s_0, a_0, s_1, a_1, \dots, s_T, a_T)$$

$$\pi_{\theta}(z) = \underbrace{p(s_0)} \prod_t \underbrace{\pi_{\theta}(a_t | s_t)} \underbrace{p(s_{t+1} | s_t, a_t)}$$

Gradient $\nabla_{\theta} J(\theta) = \nabla_{\theta} \mathbb{E}_{z \sim \pi_{\theta}} r(z) =$

$$= \nabla_{\theta} \int \pi_{\theta}(z) r(z) dz =$$
$$\times \frac{\pi_{\theta}(z)}{\pi_{\theta}(z)}$$
$$= \int \nabla_{\theta} \pi_{\theta}(z) r(z) dz =$$

$$\frac{\nabla_{\theta} \pi_{\theta}(z)}{\pi_{\theta}(z)} = \nabla_{\theta} (\log \pi_{\theta}(z))$$
$$= \int \nabla_{\theta} \log \pi_{\theta}(z) r(z) \pi_{\theta}(z) dz =$$
$$= \mathbb{E}_{z \sim \pi_{\theta}} \nabla_{\theta} \log \pi_{\theta}(z) \cdot r(z)$$

Monte Carlo estimate $\rightarrow$ From samples

$$\nabla_{\theta} J(\theta) \approx \frac{1}{N} \left[\sum_{i=1}^N \boxed{r(z_i)} \cdot \nabla_{\theta} \log \pi_{\theta}(z_i) \right]$$

Not differentiable!

1.1. Improvements

Ⓐ high variance

⇒ Increase signal: this action is better than typical

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau} (r(\tau) - \underbrace{b}_{\text{baseline}}) \cdot \sum_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$$

Ⓑ Reward-to-go

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau} r(\tau) \times \sum_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$$

entire trajectory reward every action

$$G_t = \sum_{t' \geq t} r_{t'}$$

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau} \left[\sum_t G_t \cdot \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right]$$

③ Discounting

$$\gamma \in [0, 1)$$

$$G_t = \sum_{t' \geq t} \gamma^{t'-t} r_{t'}$$

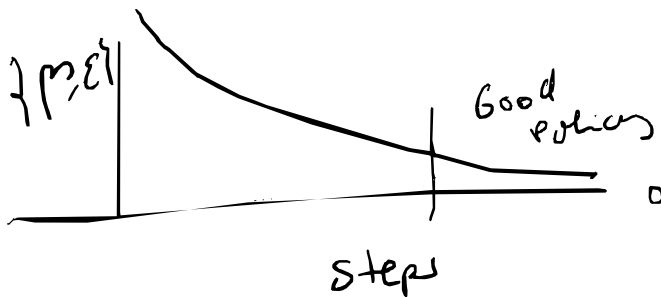
④ Exploration

④.1 Entropy Regularization

$$J(\theta) = \underbrace{\mathbb{E}_{\tau} r(\tau)}_{\text{reward}} + \underbrace{\beta \mathbb{E}_s H(\pi_{\theta}(\cdot|s))}_{\text{uncertain policies}}$$

↑
entropy

↑
anneal



④.2 ϵ -greedy

② ACTOR-CRITIC

$$V^{\pi}(s)$$

value : expect reward from state s
under policy π

Actor : $\pi_{\theta}(a|s) \sim \text{policy}$

Critic : $V_{\phi}^{\pi}(s)$

↓ Trained to predict the actual reward

$$\mathbb{E} (V_{\phi}(s_t) - G_t)^2$$

$$\nabla_{\theta} J_{\theta} = \mathbb{E} \left[\sum_t \underbrace{\hat{A}_t}_{\text{Advantage}} \nabla_{\theta} \log \pi_{\theta}(a_t|s_t) \right]$$

$$(G_t - \underbrace{V_{\phi}^{\pi}(s_t)}_{\text{baseline}})$$

baseline
(state dependent)

③ Q-LEARNING

What if we know the value of a given action?

$$Q^{\pi}(s, a) = \frac{E}{\tau} \sum_{t' \geq t} \left[\underbrace{r_{t'} | s_t = s, a_t = a, \pi}_{\text{Take action } a \text{ on state } s, \text{ and afterward we follow the policy } \pi} \right]$$

action-value function

Optimal Policy

$$\pi^*(s) = \arg \max_a \underbrace{Q^*(s, a)}$$

Bellman - Optimality Relation

$$Q^*(s, a) = \frac{E}{s'} \left\{ \underbrace{r}_{\text{immediate reward}} + \max_{a'} \underbrace{Q^*(s', a')}_{2} \right\}$$

Loss

$$L(\phi) = E \left[(r + \max_{a'} Q_{\phi}(s', a') - Q_{\phi}(s, a))^2 \right]$$